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An Internship Report
On

“FAKE NEWS DETECTION”

Submitted in Partial Fulfillment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

Submitted By

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Carried out at

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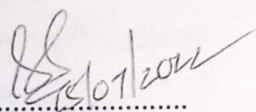
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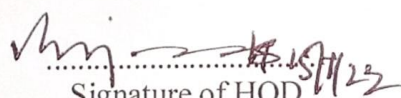
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CERTIFICATE

This is to certify that the Internship work entitled **"FAKE NEWS DETECTION"** carried out by **SONIYA C J** bearing **USN:1SJ18CS098** a bonafide student of **Sri Jagadguru Chandrashekaranaatha Institute of Technology** in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering** of **Visvesvaraya Technological University, Belgaum** during the year **2021-22**. It is certificated that all corrections suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The Internship report has been approved as it satisfies the academic requirements in respect of Internship work prescribed for the said Degree.


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- 2.

COMPANY CERTIFICATE



Pre-Incubated at



CERTIFICATE OF COMPLETION

This certifies that

Soniya C J

has completed one month Internship on

Artificial Intelligence & Machine Learning

from 1st September to 30th September 2021 at Tequed Labs

and has worked on a Project Titled

"Fake News Detection"

USN: 1SJ18CS098

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Internship ID: TLS21A175

Supreeth Y S, CEO

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DECLARATION

I, **SONIYA C J**, student of VIII semester B.E in Computer science & Engineering at S J C Institute of Technology, Chickballapur, hereby declare that the Internship work entitled “FAKE NEWS DETECTION“ has been independently carried out by me under the supervision of **Shrihari M R**, Assistant Professor, and the coordinator **Narendra Babu C and Swetha T** Assistant Professors, submitted in partial fulfillment of the course requirement for the award of degree in **Bachelor of Engineering in Computer Science & Engineering** of **Visveswaraya Technological University, Belgavi** during the year 2021-2022. I further declare that the report has not been submitted to any other University for the award of any other degree.

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ABSTRACT

In our modern era where the internet is global, everyone relies on various online resources for news. Along with the increase in the use of social media platforms like Facebook, Twitter, etc. news spread rapidly among millions of users within a very short span of time. The spread of fake news has far-reaching consequences like the creation of biased opinions. The project demonstrated for detecting the fake news. The dataset was provided by the company. Here I am performing binary classification of various news articles available online with the help of concepts pertaining to Artificial Intelligence, Natural Language Processing and Machine Learning. Using decision tree classifier provides the ability to classify the news as fake or real.

ACKNOWLEDGEMENT

With reverential pranam, we express my sincere gratitude and salutations to the feet of his holiness **Byravaikya Padmabhushana Sri Sri Sri Dr. Balagangadharanatha Maha Swamiji**, & his holiness **Jagadguru Sri Sri Sri Dr. Nirmalanandanatha Swamiji** of Sri Adichunchanagiri Mutt for their unlimited blessings. First and foremost we wish to express my deep sincere feelings of gratitude to our institution, **Sri Jagadguru Chandrashekarana Swamiji Institute of Technology**. For providing me an opportunities for completing my internship work successfully.

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SONIYA C J(1SJ18CS098)

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CHAPTER - 1

COMPANY PROFILE

1.1 History of the Organization

Tequed Labs Private Limited is a Private incorporated on 22 January 2018. It is classified as Non-govt Company and is registered at Registrar of Companies, Bangalore. Tequed Labs is a research and development centre and educational institute based in Bangalore. They are focused on providing quality education on latest technologies and develop products which are of great need to the society. They also involve in distribution and sales of latest electronic innovation products developed all over the globe to their customers. They run a project consultancy where they undertake various projects from wide range of companies and assist them technically and build products and provide services to them. They are continuously involved in research about futuristic technologies and finding ways to simplify them for their clients.

1.1.1 Objectives

- To be a world-class research and development organization committed to enhancing stakeholder's value.
- To build best products that is socially innovative with high-quality attributes and provides excellent education to all.
- Zeal to excel and zest for change. Respect for dignity and potential of individuals.
- They are continuously involved in research about futuristic technologies and finding ways to simplify them for their clients.

1.1.2 Operations of the Organization

- The organization is focused on providing quality education on latest technologies and develop products which are of great need to the society.
- They also involve in distribution and sales of latest electronic innovation products developed all over the globe to their customers.
- They run a project consultancy where they undertake various projects from wide range of companies and assist them technically and build products and provide services to them.

- They are continuously involved in research about futuristic technologies and finding ways to simplify them for their clients.

1.2 Major Milestones

Tequed Labs is a reliable organization engaged in mca provider a qualitative range of industrial products. They are also one of the leading companies of this highly commendable range of products. The team of experts maintains a vigil on the quality of the products. Every single piece of work is ensured with proper quality assurance. Since the inception in 22/01/2018, they are continually improving our quality to serve their clients better. Use of modern technology, industry standards, timely and quality deliveries, experienced workforce are their USPs.

In today's competitive marketplace, it is important to bring the businesses and technologies together to deliver on your promise. More than ever, Tequed Labs is committed to deliver on our promise so that you can deliver on yours, the success of your organization.

1.3 Structure of the Organization

It is classified as Non-govt Company and is registered at Registrar of Companies, Bangalore. Tequed Labs is a research and development center and educational institute based in Bangalore. They are focused on providing quality education on latest technologies and develop products which are of great need to the society. They also involve in distribution and sales of latest electronic innovation products developed all over the globe to their customers. The intern is honored by the internship program under this curriculum. This program has enhanced the skill and enthusiasms of the students as they get knowledge of the company environments and to learn different aspects of working mechanism that prevail in the organizations.

Through the years, and have been successfully delivering value to their customers. They truly believe that their customer's success is company success. Company don't look at themselves as a vendor for their projects instead, people would be excited to hear some of their stories and know to what extent company have gone in the interest of the success of their customers and they work hard to make that happen.

1.4 Services Offered

- Trained students will avail premium job recommendation from Job Square's Super Match feature.
- Offers training on trending technologies such as Cyber Security, Full Stack Web Development, Internet of Things, Artificial Intelligence and Machine Learning.
- Provides flexible learning platform, with expert tutors to explain the trending technologies to the interns.
- Provides certifications once the interns are trained and successfully completed the projects assigned to them.

CHAPTER-2

ABOUT THE DEPARTMENT

2.1 Specific Functionalities of the Department

There are several departments in the organization. The Tequed Labs provides online courses related to IT technology, Aptitude. In IT sector, it offers several services which includes Cloud Computing, Cyber Security, Full Stack Web Development, Internet of Things, Artificial Intelligence and Machine Learning. It also provides additional technical courses such as Data Structures, Java, Python programming languages, MongoDB, Bug Bounty, Design and Analysis of programs, Robotic process automation, Programming in C++ and many more courses available to get a hand on experience on trending technologies at an affordable price.

They are focused on providing quality education on latest technologies and develop products which are of great need to the society. They also involve in distribution and sales of latest electronic innovation products developed all over the globe to their customers. The intern is honored by the internship program under this curriculum. This program has enhanced the skill and enthusiasms of the students as they get knowledge of the company environments and to learn different aspects of working mechanism that prevail in the organizations. They are continually improving our quality to serve their clients better. Use of modern technology, industry standards, timely and quality deliveries, experienced workforce are their USPs.

2.2 Roles and Responsibilities of Individuals

Since the internship was online, to ensure easy onboarding of interns, the company had additional individuals who took care of the smooth run of online training.

- Operation and Strategy Head- Ensured there were no difficulties for interns while onboarding. Best of mentors and doubt clarifying sessions were arranged too.
- Technical Lead- Ensured the technicalities of online training to be smooth. Best platforms were arranged for our meetings and trainings.
- Mentors- They have helped us to understand the concepts, gave us tasks to get practical take a way and clarified doubts to the best.

- Interns- Worked through the tasks given either individually or in a group

2.3 Testing

Testing was done according to the Corporate Standards. As each component was being built, Unit testing was performed in order to check if the desired functionality is obtained. Each component in turn is tested with multiple test cases to verify if it is properly working. These unit tested components are integrated with the existing built components and then integration testing is performed. Here again, multiple test cases are run to ensure the newly built component runs in co-ordination with the existing components. Unit and Integration testing are iteratively performed until the complete product is built.

Once the complete product is built, it is again tested against multiple test cases and all the functionalities. The product could be working fine in the developer's environment but might not necessarily work well in all other environments that the users could be using. Hence, the product is also tested under multiple environments (Various operating systems and devices). At every step, if a flaw is observed, the component is rebuilt to fix the bugs. This way, testing is done hierarchically and iteratively.

CHAPTER-3

TASK PERFORMED

Training Program: The internship is a platform where the trainees are assigned with the specific task. In the initial days of the internship, I was trained on the following:

- Python Programming
- Machine Learning Algorithms

A. Pre-processing Data:

Social media data is highly unstructured – majority of them are informal communication with typos, slangs and bad-grammar etc. Quest for increased performance and reliability has made it imperative to develop techniques for utilization of resources to make informed decisions. To achieve better insights, it is necessary to clean the data before it can be used for predictive modeling. For this purpose, basic pre-processing was done on the News training data. This step was comprised of

Data Cleaning:

While reading data, we get data in the structured or unstructured format. A structured format has a well-defined pattern whereas unstructured data has no proper structure. In between the 2 structures, we have a semi-structured format which is a comparably better structured than unstructured format.

Cleaning up the text data is necessary to highlight attributes that we're going to want our machine learning system to pick up on. Cleaning (or pre-processing) the data typically consists of a number of steps:

a) Remove punctuation

Punctuation can provide grammatical context to a sentence which supports our understanding. But for our vectorizer which counts the number of words and not the context, it does not add value, so we remove all special characters. eg: How are you?->How are you

b) Tokenization

Tokenizing separates text into units such as sentences or words. It gives structure to previously unstructured text. eg: Plata o Plomo-> 'Plata', 'o', 'Plomo'.

c) Remove stopwords

Stopwords are common words that will likely appear in any text. They don't tell us much about our data so we remove them. eg: silver or lead is fine for me-> silver, lead, fine.

d) Stemming

Stemming helps reduce a word to its stem form. It often makes sense to treat related words in the same way. It removes suffixes, like "ing", "ly", "s", etc. by a simple rule-based approach. It reduces the corpus of words but often the actual words get neglected. eg: Entitling, Entitled -> Entitle. Note: Some search engines treat words with the same stem as synonyms.

B. Feature Generation:

We can use text data to generate a number of features like word count, frequency of large words, frequency of unique words, n-grams etc. By creating a representation of words that capture their meanings, semantic relationships, and numerous types of context they are used in, we can enable computer to understand text and perform Clustering, Classification etc.

Vectorizing Data: Vectorizing is the process of encoding text as integers i.e. numeric form to create feature vectors so that machine learning algorithms can understand our data.

1. Vectorizing Data: Bag-Of-Words Bag of Words (BoW) or CountVectorizer describes the presence of words within the text data. It gives a result of 1 if present in the sentence and 0 if not present. It, therefore, creates a bag of words with a document-matrix count in each text document.

2. Vectorizing Data: N-Grams N-grams are simply all combinations of adjacent words or letters of length n that we can find in our source text. Ngrams with n=1 are called unigrams. Similarly, bigrams (n=2), trigrams (n=3) and so on can also be used. Unigrams usually don't contain much information as compared to bigrams and trigrams. The basic principle behind n-grams is that they capture the letter or word is likely to follow the given word. The longer the n-gram (higher n), the more context you have to work with.

3. Vectorizing Data: TF-IDF It computes "relative frequency" that a word appears in a document compared to its frequency across all documents TF-IDF weight represents the relative importance of a term in the document and entire corpus. TF stands for Term Frequency: It calculates how frequently a term appears in a document. Since, every document size varies, a term may appear more in a long sized document than a short one. Thus, the length of the document often divides Term frequency.

C. Algorithms used for Classification**➤ Naïve Bayes Classifier:**

This classification technique is based on Bayes theorem, which assumes that the presence of a particular feature in a class is independent of the presence of any other feature. It provides way for calculating the posterior probability.

➤ Random Forest:

Random Forest is a trademark term for an ensemble of decision trees. In Random Forest, we've collection of decision trees (so known as "Forest"). To classify a new object based on attributes, each tree gives a classification and we say the tree "votes" for that class. The forest chooses the classification having the most votes (over all the trees in the forest). The random forest is a classification algorithm consisting of many decisions trees. It uses bagging and feature randomness when building each individual tree to try to create an uncorrelated forest of trees whose prediction by committee is more accurate than that of any individual tree. Random forest, like its name implies, consists of a large number of individual decision trees that operate as an ensemble. Each individual tree in the random forest spits out a class prediction and the class with the most votes becomes our model's prediction. The reason that the random forest model works so well is: A large number of relatively uncorrelated models (trees) operating as a committee will outperform any of the individual constituent models.

➤ Logistic Regression:

It is a classification not a regression algorithm. It is used to estimate discrete values (Binary values like 0/1, yes/no, true/false) based on given set of independent variable(s). In simple words, it predicts the probability of occurrence of an event by fitting data to a logit function. Hence, it is also known as logit regression. Since, it predicts the probability, its output values lies between 0 and 1 (as expected). Mathematically, the log odds of the outcome are modeled as a linear combination of the predictor variables.

➤ Passive Aggressive Classifier:

The Passive Aggressive Algorithm is an online algorithm; ideal for classifying massive streams of data (e.g. twitter). It is easy to implement and very fast. It works by taking an example, learning from it and then throwing it away. Such an algorithm remains passive for a correct classification outcome, and turns aggressive in the event of a miscalculation, updating and adjusting. Unlike most other algorithms, it does not converge. Its purpose is to make updates that correct the loss, causing very little change in the norm of the weight vector.

CHAPTER-4

REFLECTION NOTES

4.1 Experience

As per our experience during the internship, Tequed Labs follows a good work culture and it has friendly employees, starting from the staff level to the management level. The trainers are well versed in their fields and they treat everyone equally. There is no distinguishing between fresher graduates and corporates and everyone is respected equally. There is a lot of teamwork followed in every task, be it hard or easy and there is a very calm and friendly atmosphere maintained at all times. There is a lot of scope for self-improvement due to the great communication and support that can be found. Interns have been treated and taught well and all our doubts and concerns regarding the training or the companies have been properly answered. All in all, Tequed Labs was a great place for a fresher to start career and also for a corporate to boost his/her career. It has been a great experience to be an intern in such a reputed organization.

4.2 Technical Outcomes

4.2.1 System Requirements and Specification

HARDWARE REQUIREMENTS:

- Processor : x86 or x64
- Hard Disk: 500 GB or more.
- Ram : 512 MB(minimum) , 1 GB(recommended)

SOFTWARE REQUIREMENTS:

Operating System : Windows or Linux

Platform used : Anaconda Navigator (Jupyter notebook)

4.3 System Analysis and Design

4.3.1 Existing System

There exists a large body of research on the topic of machine learning methods for deception detection, most of it has been focusing on classifying online reviews and publicly available social media posts. Particularly since late 2016 during the American Presidential election, the

question of determining ‘fake news’ has also been the subject of particular attention within the literature.

Conroy, Rubin, and Chen outlines several approaches that seem promising towards the aim of perfectly classify the misleading articles. They note that simple content-related n-grams and shallow parts-of-speech (POS) tagging have proven insufficient for the classification task, often failing to account for important context information. Rather, these methods have been shown useful only in tandem with more complex methods of analysis. Deep Syntax analysis using Probabilistic Context Free Grammars (PCFG) have been shown to be particularly valuable in combination with n-gram methods. Feng, Banerjee, and Choi are able to achieve 85%-91% accuracy in deception related classification tasks using online review corpora.

Feng and Hirst implemented a semantic analysis looking at ‘object:descriptor’ pairs for contradictions with the text on top of Feng’s initial deep syntax model for additional improvement. Rubin, Lukoianova and Tatiana analyze rhetorical structure using a vector space model with similar success. Ciampaglia et al. employ language pattern similarity networks requiring a pre-existing knowledge base.

4.3.2 Disadvantages of the Existing System

- Information was not clear and not able to extract the correct information in the bulk of news.
- Defamation is among the disadvantages of fake news
- False Perception
- Fake News may lead to Social Unrest

4.3.3 Proposed System

Model is build based on the count vectorizer or a tfidf matrix (i.e) word tallies relatives to how often they are used in other articles in your dataset) can help . Since this problem is a kind of text classification, Implementing a Naive Bayes classifier will be best as this is standard for text-based processing. The actual goal is in developing a model which was the text transformation (count vectorizer vs tfidf vectorizer) and choosing which type of text to use (headlines vs full text). Now the next step is to extract the most optimal features for countvectorizer or tfidf-vectorizer, this is done by using a n-number of the most used words, and/or phrases, lower casing or not, mainly removing the stop words which are common words such as “the”, “when”, and “there” and only using those words that appear at least a given number of times in a given text dataset.

4.3.4 Advantages of the Proposed System

- Information was very clear and understandable.
- It gives accurate predictions which is very clear to the user.
- User friendly and faster time compatibility.

4.4 System Architecture

4.4.1 Data flow diagram

The DFD takes an input-process-output view of a system i.e., data objects flow into the software, are transformed by processing elements, and resultant data objects flow out of the software. The dataset contains real and fake news information. Then the information is fed to algorithm .Thus news is analyzed as fake or real.

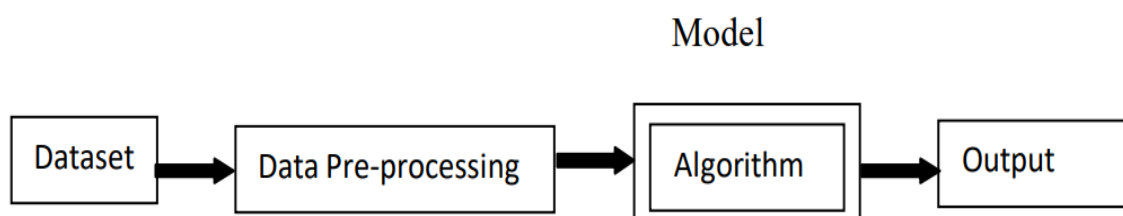


Figure 4.1 : Data Flow Diagram

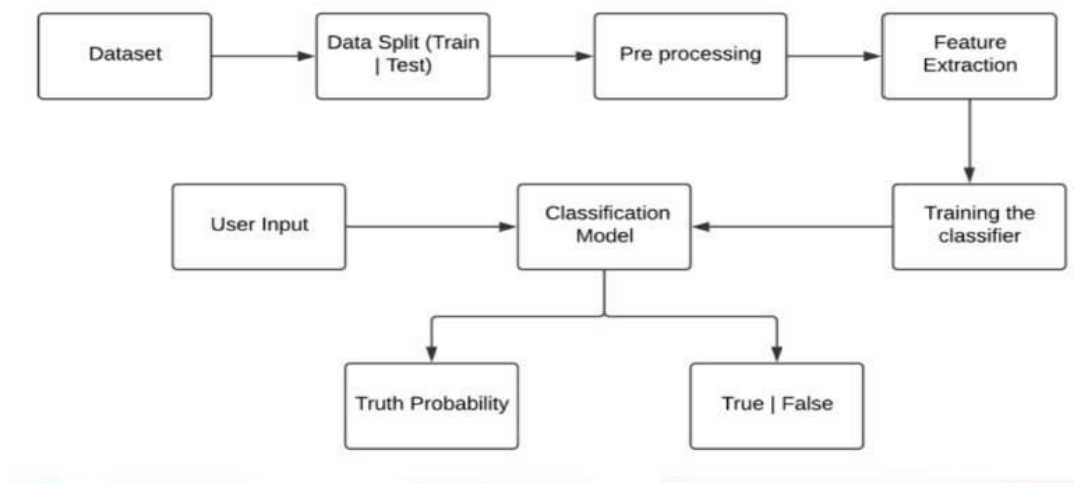


Figure 4.2 : System architecture

4.5 Implementation

A. Static Search Implementation-

In static part, we have trained and used 3 out of 4 algorithms for classification. They are Naïve Bayes, Random Forest and Logistic Regression.

Step 1: In first step, extracting features from the already pre-processed dataset. These features are; Bag-of-words, Tf-Idf Features and N-grams.

Step 2: Here, built all the classifiers for predicting the fake news detection. The extracted features are fed into different classifiers. We have used Naive-bayes, Logistic Regression, and Random forest classifiers from sklearn. Each of the extracted features was used in all of the classifiers.

Step 3: Once fitting the model, comparing the f1 score and checked the confusion matrix.

Step 4: After fitting all the classifiers, 2 best performing models were selected as candidate models for fake news classification.

Step 5: Now performed parameter tuning by implementing GridSearchCV methods on these candidate models and chosen best performing parameters for these classifier.

Step 6: Finally selected model was used for fake news detection with the probability of truth.

Step 7: Our finally selected and best performing classifier was Logistic Regression which was then saved on disk. It will be used to classify the fake news.

It takes a news article as input from user then model is used for final classification output

that is shown to user along with probability of truth.

B. Dynamic Search Implementation-

Our dynamic implementation contains 3 search fields which are-

- 1) Search by article content.
- 2) Search using key terms.
- 3) Search for website in database.

In the first search field we have used Natural Language Processing for the first search field to come up with a proper solution for the problem, and hence we have attempted to create a model which can classify fake news according to the terms used in the newspaper articles. Our application uses NLP techniques like Count Vectorization and TF-IDF Vectorization before passing it through a Passive Aggressive Classifier to output the authenticity as a percentage probability of an article.

The second search field of the site asks for specific keywords to be searched on the net upon which it provides a suitable output for the percentage probability of that term actually being present in an article or a similar article with those keyword references in it.

The third search field of the site accepts a specific website domain name upon which the implementation looks for the site in our true sites database or the blacklisted sites database. The true sites database holds the domain names which regularly provide proper and authentic news and vice versa. If the site isn't found in either of the databases then the implementation doesn't classify the domain it simply states that the news aggregator does not exist.

Working-

The problem can be broken down into 3 statements-

- 1) Use NLP to check the authenticity of a news article.
- 2) If the user has a query about the authenticity of a search query then we he/she can directly search on our platform and using our custom algorithm we output a confidence score.
- 3) Check the authenticity of a news source.

These sections have been produced as search fields to take inputs in 3 different forms in our implementation of the problem statement.

4.6 Screen Shots

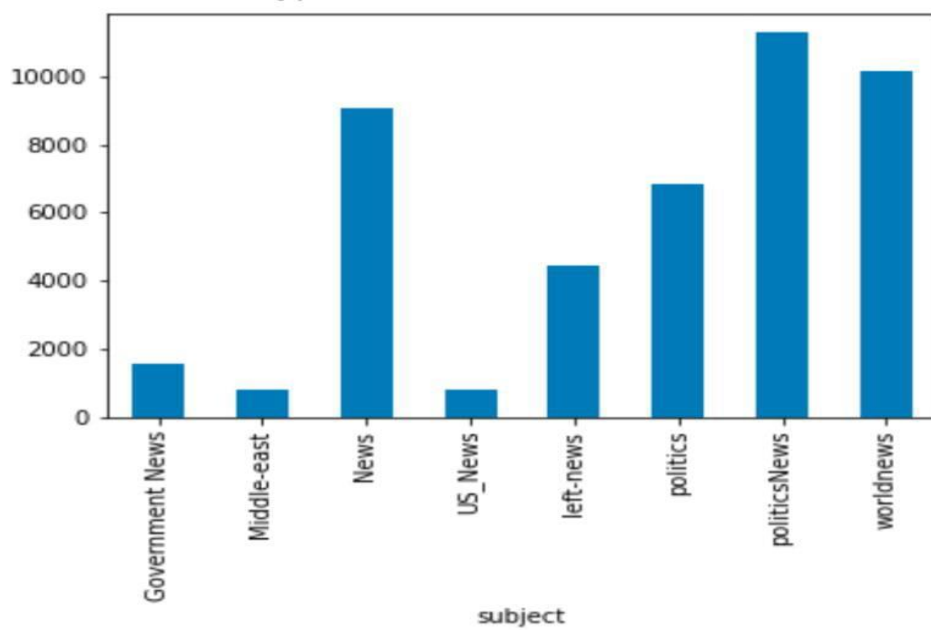


Figure 4.3 : Information related to subject present in the dataset.

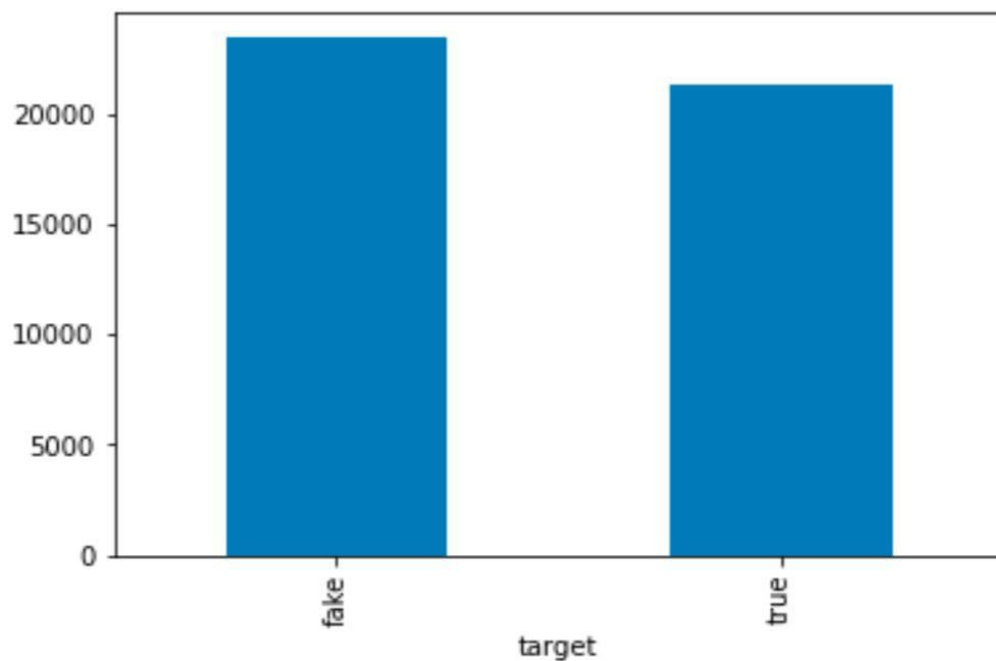


Figure 4.4 : Analysing fake and real news from the dataset.

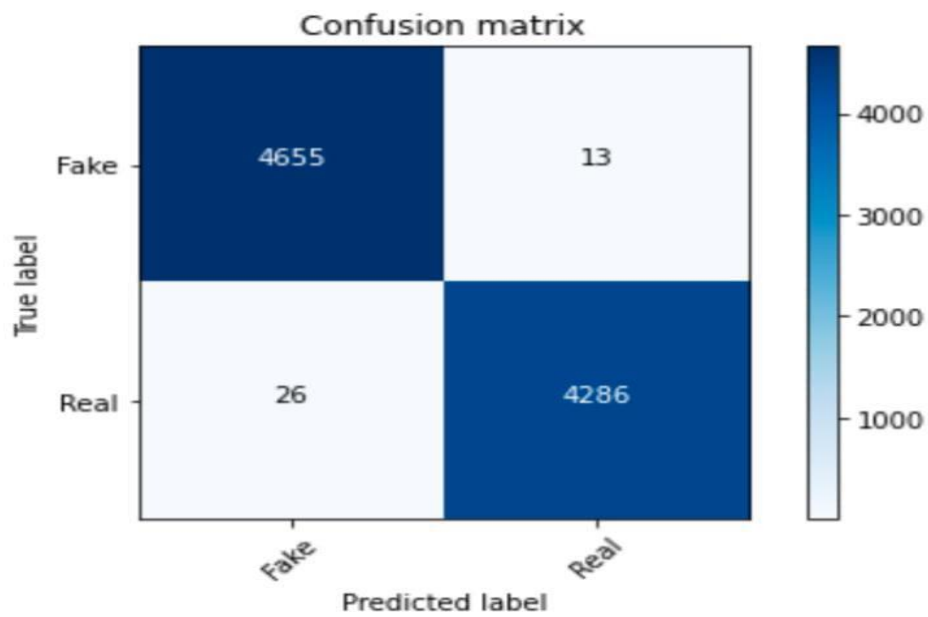


Figure 4.5 : Confusion matrix

CHAPTER-5

CONCLUSION

The task of classifying news manually requires in-depth knowledge of the domain and expertise to identify anomalies in the text. The data used in work contains news articles from various domains to cover most of the news rather than specifically classifying political news. The primary aim of the research is to identify patterns in text that differentiate fake articles from true news. Here I extracted different textual features from the articles and used the feature set as an input to the models. The learning models were trained and parameter-tuned to obtain optimal accuracy.

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